

# Jay Unadkat

MSc Information Technology

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## Education

**Dhirubhai Ambani University (formerly DA-IICT)**

MSc Information Technology – CPI: 8.00/10

2025 – Present

Gandhinagar, Gujarat

**GLS University**

BCA | CPI: 8.40/10

2022 – 2025

Ahmedabad, Gujarat

## Technical Skills

**Languages:** Python, C++, Java, JavaScript, SQL

**AI/LLM:** LangChain, LangGraph, RAG, AI Agents, Tool Calling, Vector Databases, OpenAI API, Hugging Face

**Frameworks:** React.js, Node.js, Express.js, Tailwind CSS, Streamlit

**Databases:** MongoDB, MySQL, PostgreSQL

**Backend/Infra:** Redis, BullMQ, REST APIs, JWT Authentication, Load Testing (Autocannon)

**Developer Tools:** Git, GitHub, Postman, VS Code, AWS

**Coursework:** Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

## Projects

**VoyAgent - Multi-Agent AI Travel Planning System** | Python, LangGraph, LangChain, FastAPI, Express.js, MongoDB



- Architected a **stateful multi-agent AI system** using **LangGraph**, orchestrating a Supervisor with specialized Planner, Flight, Hotel, Itinerary, Critic, and Response agents through graph-based workflows to generate personalized travel plans.
- Designed an **Agentic AI workflow** incorporating **Human-in-the-Loop (HITL)** checkpoints and a Critic Agent that validates budget, destination, and travel constraints, interrupting execution to collect user clarification before resuming the workflow.
- Engineered a highly concurrent execution pipeline using **fan-out/fan-in parallel orchestration**, enabling hotel, flight, and itinerary agents to execute simultaneously while maintaining shared conversational state and reducing end-to-end workflow latency.
- Leveraged **Structured Outputs**, **Pydantic schemas**, and **SerpAPI** to generate strongly typed travel recommendations grounded in real-time flight and hotel information, significantly reducing pricing and availability hallucinations.
- Built a production-ready full-stack platform using **Next.js**, **Express.js**, **FastAPI**, and **MongoDB** featuring JWT authentication, persistent multi-turn conversations, REST APIs, and modular AI workflow orchestration.

**CodeStage - Online Code Evaluation Platform** | React.js, Node.js, Express.js, MongoDB, Redis, BullMQ, Judge0 API



- Architected and deployed a full-stack coding platform enabling automated multi-test-case evaluation, with a decoupled backend supporting concurrent submission processing.
- Engineered an asynchronous, event-driven execution pipeline using **Redis** as a message broker and **BullMQ** for distributed job queueing, decoupling submission intake from code execution via horizontally scalable worker processes.
- Validated pipeline reliability under **concurrent** load via custom load-testing, achieving a **100% success rate** with zero dropped requests across **1000+ simultaneous** submission requests.
- Integrated the **Judge0 API** for sandboxed, isolated code execution, capturing compile/runtime diagnostics, execution time, and memory usage to deliver granular test-case-level feedback.
- Implemented a real-time status-polling mechanism on the frontend to reflect **asynchronous** job-lifecycle transitions (Queued → Processing → Completed), eliminating unresponsive UI states during evaluation.

## Achievements

- Secured 1st Runner-Up at HackOut'25 (Synapse, DA-IICT) among 1000+ teams, leading the team during the hackathon.
- Secured 2nd Runner-Up in FaceOff, a competitive 1v1 DSA/CP contest conducted by MSTC (Microsoft Student Technical Club).
- Solved 500+ Data Structures and Algorithms problems across LeetCode, Codeforces, and GeeksforGeeks. LeetCode Max Rating: 1808.